A logarithmic endpoint partial result for the continuous Bennett–Carl conjecture

Run `fa_banach_001`, lane 13

21 July 2026

Status. Candidate partial result, likely valid. The $(p,1)$-summing conjecture itself remains open.

1 Source problem

Astashkin, Leśnik and Wojciechowski [1] define, for $1 < p < 2$,

$$X_p = \Lambda_{p,W_p}, \qquad W_p(t) = \log^{1-p}(e/t),$$

and prove that the inclusion $J : X_p \to L^p[0,1]$ is $(q,1)$-absolutely summing for every $p < q < 2$. Their Conjecture 3.4 asks whether J is $(p,1)$-absolutely summing.

Thus one could also conclude that $\mathcal{T} : Y \to L^p(X_p)$, if there holds

$$Y \subset (L^\infty(L^\infty), L^\infty(G))_{\theta,p}. \tag{5}$$

Regardless boundedness of $\mathcal{T}$, the question remains if Theorem 1.1 holds true with $q = p$.

Conjecture 3.4. *Let $1 < p < 2$. Then the inclusion $X_p \subset L^p$ is $(p,1)$-absolutely summing.*

The next trivial example explains that the choice of L^p in the conjecture is optimal, i.e. L^p cannot be replaced by any smaller Lebesgue space.

Example 3.5. *For $1 < p < q$ the inclusion $X_p \subset L^q$ is not $(p,1)$-summing if $q > p$.*

Proof. Let $f_i^k = \chi_{A_i^k}$, where $A_i^k = \left[\frac{i-1}{k}, \frac{i}{k}\right)$. Then

The crop is from page 12 of the source PDF. In finite-family form the conjecture asks for a constant C_p such that

$$\left(\sum_i \|f_i\|_p^p\right)^{1/p} \le C_p \max_{\varepsilon_i=\pm 1} \left\|\sum_i \varepsilon_i f_i\right\|_{X_p}. \tag{1}$$

For a Young function Φ, write ℓ_Φ for the Luxemburg sequence space. We say that $T : E \to F$ is $(\Phi,1)$-summing if

$$\left\|(\|Tx_i\|_F)_i\right\|_{\ell_\Phi} \le C \sup_{x^*\in B_{E^*}} \sum_i |x^*(x_i)|$$

for all finite families. The quantity on the right equals $\max_{\varepsilon_i=\pm1} \|\sum_i \varepsilon_i x_i\|_E$.

2 Proof intuition

The source proof loses the endpoint only in the embedding of the exponential Orlicz space $\operatorname{Exp} L^{p'}$ into X_q. Its norm can be computed from rearrangements: the q-th power grows like $(q-p)^{-1}$. All other constants can be kept uniform while $q \downarrow p$. Consequently

$$\pi_{q,1}(J)^q \lesssim_p (q-p)^{-1}.$$

Integrating this family of estimates with weight $(q-p)^{\beta-1}$ turns u^{p+q-p} into $u^p/\log^\beta(e/u)$. This produces a single logarithmic endpoint space for every $\beta > 1$. The integral diverges at $\beta = 1$, so the mechanism does not prove the original conjecture.

3 The partial theorem

For $\beta > 1$, let Φ_β be any Young function such that

$$\Phi_\beta(u) \asymp_{p,\beta} \frac{u^p}{\log^\beta(e/u)} \qquad (0 < u \le u_0). \tag{2}$$

Such a Young function is obtained by convexifying the displayed function near zero and continuing it convexly; equivalent choices give the same sequence space up to equivalent norms.

Theorem 1. *Let $1 < p < 2$ and $\beta > 1$. The inclusion $J : X_p \to L^p[0,1]$ is $(\Phi_\beta, 1)$-absolutely summing. Thus there is $C_{p,\beta} < \infty$ such that every finite family $(f_i) \subset X_p$ obeys*

$$\left\| (\|f_i\|_p)_i \right\|_{\ell_{\Phi_\beta}} \le C_{p,\beta} \max_{\varepsilon_i = \pm 1} \left\| \sum_i \varepsilon_i f_i \right\|_{X_p}. \tag{3}$$

Moreover,

$$\ell^p \subsetneq \ell_{\Phi_\beta} \subsetneq \ell^q \quad \textit{for every } q > p.$$

In particular, (3) *is a genuine endpoint-scale strengthening of the separate* $(q,1)$ *estimates in [1], but it is weaker than* (1).

4 Proof

We first quantify the source argument. Put $p' = p/(p-1)$ and $E = \operatorname{Exp} L^{p'}$.

Lemma 1 (Uniform Rademacher lower estimate)**.** *There exist $\eta > 0$ and $c_p > 0$ such that, for every $q \in (p, p+\eta)$ and every finite scalar family (a_i),*

$$\left\| \sum_i a_i r_i \right\|_{X_q} \ge c_p \left(\sum_i |a_i|^q \right)^{1/q}.$$

Proof. Reorder so that $|a_1| \ge \cdots \ge |a_n|$ and set $A_k = \sum_{i \le k} |a_i|$. The first k signs align with the coefficients with probability 2^{-k}. Conditional on this event, the remaining signed sum is symmetric and is nonnegative with probability at least $1/2$. Therefore

$$\left(\sum_i a_i r_i \right)^* (2^{-k-1}) \ge A_k.$$

Writing $W_q(t) = \log^{1-q}(e/t)$ and integrating on the adjacent dyadic intervals gives

$$\Big\|\sum_i a_i r_i\Big\|_{X_q}^q \geq \sum_{k=1}^n A_k^q\big[W_q(2^{-k-1}) - W_q(2^{-k-2})\big].$$

By the mean value theorem, uniformly for q in a sufficiently small compact interval above p, the bracket is at least $c_p^q(k+2)^{-q}$. Since $A_k/k \geq |a_k|$, the last display is at least $c_p^q \sum_k |a_k|^q$, after changing c_p by a fixed factor. □

Lemma 2 (Quantitative near-endpoint estimate)**.** *After possibly decreasing η, for $p < q < p + \eta$,*

$$\pi_{q,1}(J) \leq C_p(q-p)^{-1/q}. \tag{4}$$

Proof. The elementary rearrangement estimate

$$\|f\|_{X_p}^p \geq f^*(t)^p W_p(t)$$

implies $f^*(t) \leq \|f\|_{X_p} \log^{1/p'}(e/t)$, and hence $X_p \hookrightarrow E$ with a constant depending only on p. Conversely, the standard rearrangement bound in E gives $f^*(t) \leq C_p\|f\|_E \log^{1/p'}(e/t)$. Since

$$dW_q(t) = \frac{q-1}{t\log^q(e/t)}\,dt,$$

we obtain

$$\begin{aligned}
\|f\|_{X_q}^q &\leq C_p^q\|f\|_E^q(q-1)\int_0^1 \frac{\log^{q/p'}(e/t)}{t\log^q(e/t)}\,dt \\
&= C_p^q\|f\|_E^q(q-1)\int_1^\infty u^{-q/p}\,du \\
&= C_p^q\|f\|_E^q\frac{p(q-1)}{q-p}.
\end{aligned}$$

Thus

$$\|E \hookrightarrow X_q\| \leq C_p(q-p)^{-1/q} \tag{5}$$

uniformly in the stated interval.

Let $(f_i) \subset X_p$. The q-concavity inequality in L^p and Lemma 1, applied pointwise, yield

$$\Big(\sum_i \|f_i\|_p^q\Big)^{1/q} \leq C_p \Big\|\Big\|\sum_i r_i(s) f_i(t)\Big\|_{X_q(s)}\Big\|_{L^p(t)}.$$

Use (5), followed by the transposition estimate

$$\mathcal{T} : L^\infty(E) \longrightarrow L^p(E)$$

from Lemma 3.1 of [1], and finally $X_p \hookrightarrow E$. We get

$$\begin{aligned}
\Big(\sum_i \|f_i\|_p^q\Big)^{1/q} &\leq C_p(q-p)^{-1/q}\Big\|\Big\|\sum_i r_i(t) f_i(s)\Big\|_{X_p(s)}\Big\|_{L^\infty(t)} \\
&= C_p(q-p)^{-1/q}\max_{\varepsilon_i=\pm1}\Big\|\sum_i \varepsilon_i f_i\Big\|_{X_p}.
\end{aligned}$$

This is (4). □

Proof of Theorem 1. Normalize

$$A := \max_{\varepsilon_i=\pm 1} \left\| \sum_i \varepsilon_i f_i \right\|_{X_p} = 1, \qquad y_i = \|f_i\|_p.$$

Lemma 2 says, for $0 < s < \eta$,

$$\sum_i y_i^{p+s} \le \frac{C_p^{p+s}}{s}. \tag{6}$$

Also $\sup_i y_i \le C_p$: prescribed signs on one coordinate can be extended to all coordinates without decreasing the norm, and $X_p \hookrightarrow L^p$.

Choose $K \ge 2C_p$ and put $z_i = y_i/K \le 1/2$. Multiply (6) by $K^{-(p+s)} s^{\beta-1}$ and integrate over $0 < s < \eta$. Enlarging K by a fixed p-dependent factor if necessary,

$$\sum_i \int_0^\eta z_i^{p+s} s^{\beta-1}\, ds \le C_{p,\beta} K^{-p} \int_0^\eta s^{\beta-2}\, ds < \infty, \tag{7}$$

where the last integral is finite exactly because $\beta > 1$.

For $0 < z \le 1/2$, put $L = \log(1/z)$. Integrating only over $0 < s < \min(\eta, L^{-1})$ shows

$$\int_0^\eta z^{p+s} s^{\beta-1}\, ds = z^p \int_0^\eta e^{-sL} s^{\beta-1}\, ds \ge c_{p,\beta,\eta} \frac{z^p}{\log^\beta(e/z)}.$$

Together with (7), this proves

$$\sum_i \Phi_\beta(y_i/K) \le C'_{p,\beta} K^{-p}.$$

Increasing $K = K(p,\beta)$ once more makes the right side at most one, which is precisely the Luxemburg estimate (3). Homogeneity restores an arbitrary value of A.

Finally, as $u \downarrow 0$, $\Phi_\beta(u) = o(u^p)$, whereas $u^q = o(\Phi_\beta(u))$ for every $q > p$. The standard characteristic-sequence tests give the asserted strict inclusions. □

5 Verification, limitations, and novelty status

- The proof isolates every dependence on $q - p$. Lemma 1 is uniform by an explicit dyadic calculation; the sole singular factor is the integral in (5).
- The extrapolation argument is written out rather than invoked as a black box. It is the same Yano-type mechanism developed in Section 8 of Kazaniecki–Pawlewicz–Wojciechowski [2].
- The argument stops at $\beta > 1$. At $\beta = 1$, the upper integral in (7) is logarithmically divergent. No assertion is made for $\beta = 1$, and the desired $\Phi(u) = u^p$ endpoint remains open.
- Exact finite-atomic searches from the preceding attempt found no ratio above one, including perturbations of the disjoint extremizers. These tests support, but do not prove, Conjecture 3.4 and are not used above.
- A bounded web/arXiv and local-corpus search on 21 July 2026 used the source title, arXiv id, author combinations, "continuous Bennett–Carl", "endpoint summing", and logarithmic Lorentz/Orlicz variants. It found no later settlement and no explicit statement of Theorem 1.

Human-review recommendation. Promote as a likely-valid partial theorem, not as a solution of Conjecture 3.4. The most important checks are the uniformity in Lemma 1, the orientation of the two mixed norms in the transposition step, and the Luxemburg normalization in the final integral.

References

[1] S. V. Astashkin, K. Leśnik, and M. Wojciechowski, *Looking for a continuous version of Bennett–Carl theorem*, arXiv:2502.07041 (2025).

[2] K. Kazaniecki, A. Pawlewicz, and M. Wojciechowski, *Factoring the Sobolev embedding operator*, arXiv:2503.20478 (2025), Section 8.